

Real-Time Fractional Tracking (R-TFT): Recursive ϕ Structure and the Sublimation of Zero

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Abstract

This paper explores the fundamental connection between recursive ϕ -structure and the metaphysical behavior of zero within dimensional emergence. We propose that zero, rather than being a fixed point of absence, functions as a recursive attractor, undergoing sublimation into higher ϕ -coherent states through recursive phase compression. We define this process via the ϕ -recursive embedding function, introducing a new interpretation of dimensional foundations, energy inversion, and coherence collapse thresholds in resonance-based systems.

1 Introduction

In conventional physics, zero is interpreted as the null point, absence of energy, position, or information. Within the Real-Time Fractional Tracking (R-TFT) framework, however, we demonstrate that zero is not absence, but the pre-recursive attractor to which all divergent systems collapse unless sustained by ϕ -coherence.

2 Dimensional Embedding and ϕ -Sublimation

Let $\phi = \frac{1+\sqrt{5}}{2} \approx 1.61803$. Define recursive dimensional embedding for any system X as:

$$\text{Dim}(X) = \frac{\ln X}{\ln \phi} \tag{1}$$

The closer $\text{Dim}(X)$ aligns with an integer, the higher the recursive resonance coherence. Zero, by contrast, has $\ln 0 \rightarrow -\infty$, implying collapse into non-recursivity.

2.1 Sublimation Path

We define the *sublimation path of zero* as the limit case of recursive inverse compression:

$$\lim_{X \rightarrow 0^+} \phi^{\text{Dim}(X)} = 0 \tag{2}$$

But this limit does not vanish passively, it recurses inward infinitely unless reflected by a boundary of coherent resonance. This introduces the concept of:

Recursive Freeze: Zero is not an endpoint, but an internal collapse vector requiring ϕ -containment to avoid loss of dimensional persistence.

3 Sublimation and Reality Boundaries

Just as mass can decay via coherence loss ($C_\phi \rightarrow 1$), so too can systems ‘invert’ into recursive nullification. The criteria for avoiding collapse:

$$C_\phi = |\text{Dim}(X) - \text{round}(\text{Dim}(X))| < \phi^{-1} \quad (3)$$

If $C_\phi \geq \phi^{-1}$, coherence fails, and the system begins sublimating toward zero.

3.1 Recursive Density Collapse

We define recursive density ρ_ϕ as:

$$\rho_\phi = \frac{1}{1 + C_\phi} \quad (4)$$

As $C_\phi \rightarrow 1$, $\rho_\phi \rightarrow 0.5$, marking the entropic decay boundary.

4 Implications

4.1 Zero as Pre-Dimensional Lattice

Zero is reframed not as nothing, but as the *pre-image of recursion*. In systems with recursive ϕ -alignment, the threat of sublimation defines their stability margin. Existence requires ongoing coherence; collapse is not sudden, but recursive.

4.2 Reversal Symmetry and Entropic Echo

If ϕ -alignment fails and a system collapses into zero, residual phase echoes may produce:

- Recursive ghost fields
- Slippage patterns (R(t) fluctuations)
- Topological freeze (no further phase embedding possible)

These may explain vacuum instability and spontaneous fluctuation origins.

5 Conclusion

Zero is no longer the default ground, it is the infinite inward slope of recursion. Systems persist only if their recursive structures avoid the nullifying attractor of incoherence. R-TFT recasts the void not as nothing, but as an unresolvable recursion unless ϕ -stabilized.

Resonance Ethics License (REL2.0)

Scientific insight without recursive ethics leads to misalignment and collapse. This paper and the R-TFT framework are protected under REL2.0, which enforces:

- **Non-weaponization:** Recursive constructs must not be applied to warfare, surveillance, or coercive fields.
- **Coherence containment:** Any derived technologies or frameworks must preserve dimensional stability and avoid ϕ -divergent behavior.
- **Transparent recursion:** All derivative works must clearly show recursive basis and modification pathways.

The pursuit of truth demands structure. Misuse fractures coherence and accelerates collapse into the recursive zero.

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>

Source Repository: <https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT>